

Q1] $f(z, v) = u(x, y) + i v(x, y)$

$u(x, y) \uparrow \uparrow v(x, y)$

$\downarrow$ * $\begin{cases} u_x = v_y & (i) \\ v_x = -u_y & (ii) \end{cases}$ *

$U(x, y) = e^{-2uv} \cos(u^2 - v^2)$
 $V(x, y) = e^{-2uv} \sin(u^2 - v^2)$

Show: (i) $\frac{\partial}{\partial x}(U) = \frac{\partial}{\partial y}(V)$

$\frac{\partial}{\partial x} U = \frac{\partial}{\partial x} (e^{-2uv}) \cos(u^2 - v^2) + e^{-2uv} \frac{\partial}{\partial x} (\cos(u^2 - v^2))$
 $= -2 \frac{\partial}{\partial x} (uv) e^{-2uv} \cos(u^2 - v^2) + e^{-2uv} (-\sin(u^2 - v^2)) \frac{\partial}{\partial x} (u^2 - v^2)$
 $= -2 e^{-2uv} \cos(u^2 - v^2) (u_x v + u v_x)$
 $= e^{-2uv} \sin(u^2 - v^2) [2u u_x - 2v v_x]$

$\frac{\partial}{\partial y} V = \frac{\partial}{\partial y} (e^{-2uv}) \sin(u^2 - v^2) + e^{-2uv} \frac{\partial}{\partial y} (\sin(u^2 - v^2))$
 $= -2 e^{-2uv} \sin(u^2 - v^2) [u_y v + u v_y]$
 $+ e^{-2uv} \cos(u^2 - v^2) [2u u_y - 2v v_y]$

Use $u_x = v_y$ | $v_x = -u_y$

$\frac{\partial}{\partial x} U = -2 e^{-2uv} \cos(u^2 - v^2) (v_y v + u (-u_y))$
 $= e^{-2uv} \sin(u^2 - v^2) [2u v_y - 2v (-u_y)]$

$= e^{-2uv} \sin(u^2 - v^2) [2u v_y + 2v u_y]$

$+ 2 e^{-2uv} \cos(u^2 - v^2) [u v_y - v v_y]$

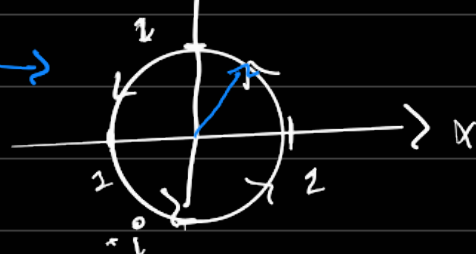
Show: $U_x = V_y$ (i)

left: to show $V_x = -U_y$

② $I_1 = \oint_{C_1} \sin[z + \pi/2] dz$ C_1

$\uparrow y$

$$z: (x, y) \mapsto (r, \theta)$$



$$z = r e^{i\theta}$$

$$r = 1$$

$$z = e^{i\theta}$$

$$dz = d e^{i\theta} = i e^{i\theta} d\theta$$

$$I_1 = \int_0^{2\pi} \frac{\sin(z + \pi/2)}{e^{i\theta}} (i e^{i\theta} d\theta)$$

$$= i \int_0^{2\pi} \sin(z + \pi/2) d\theta$$

$$* \sin(z + \pi/2) \mapsto \sin(x + \pi/2) = \cos(x)$$

$$x = 0$$

$$\sin(\pi/2) = 1 \quad \cos(0) = 1$$

$$x = \pi/2:$$

$$\sin(\pi) = 0 \quad \cos(\pi) = -1$$

$$\cos(z)$$

$$= i \int_0^{2\pi} \cos(z) d\theta$$

$$\cos(z) = \sum_{n=0}^{\infty} \frac{z^{2n} (-1)^n}{(2n)!}$$

$$= i \int_0^{2\pi} \sum_{n=0}^{\infty} \frac{z^{2n} (-1)^n}{(2n)!} d\theta = i \int_0^{2\pi} \sum_{n=0}^{\infty} \frac{(e^{i\theta})^{2n} (-1)^n}{(2n)!} d\theta$$

$$* = i \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \int_0^{2\pi} e^{i2n\theta} d\theta$$

$$= i \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left[\frac{1}{i2n} e^{i2n\theta} \right]_0^{2\pi}$$

Split $n=0$ & $n>0$ terms

$$= i \left[\int_0^{2\pi} 1 d\theta + \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} \int_0^{2\pi} e^{i2n\theta} d\theta \right]$$

$$= i \left[\int_0^{2\pi} 1 d\theta + \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} \frac{1}{i2n} e^{i2n\theta} \right]_0^{2\pi}$$

$$= i \left[2\pi + \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} \left[\frac{1}{2n} e^{i2n\theta} \right]_0^{2\pi} \right]$$

$$e^{4\pi n i} = \cos(4\pi n) + i \sin(4\pi n) = 1$$

1 0

$$e^{4\pi n i} = 1$$

$$[1 - 1] = 0$$

$$I_1 = \oint_C \frac{\sin(z + \pi/2)}{z} dz = 2\pi i$$

$$2b) I_2 = \oint_{C_2} \frac{\sin(z)}{z(z - \pi/2)} dz$$

$$z = re^{i\theta} = \pi e^{i\theta}$$

$$dz = i\pi e^{i\theta} d\theta$$

$$* \frac{1}{(z-\alpha)(z-\beta)} \mapsto \text{decompose first}$$

Partial fraction decomp:

$$\frac{1}{z(z - \pi/2)} = \frac{A}{z} + \frac{B}{(z - \pi/2)}$$

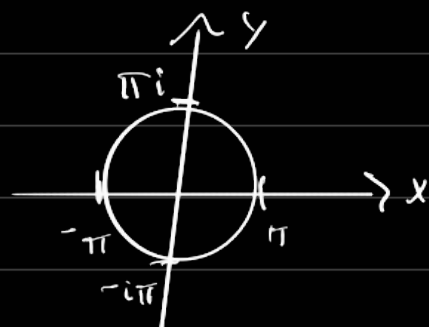
$$1 = A(z - \pi/2) + Bz$$

$$= z(A+B) - \frac{\pi}{2}A$$

LHS: $\frac{1}{\text{No } z} \mapsto \begin{matrix} -\frac{\pi}{2}A = 1 \\ A+B = 0 \end{matrix} *$

$$* \begin{cases} A = -\frac{2}{\pi} \\ B = \frac{2}{\pi} \end{cases} \Rightarrow -\frac{2}{\pi} + B = 0 \text{ or}$$

$$\frac{1}{z(z - \pi/2)} = \frac{2}{\pi} \left[\frac{1}{(z - \pi/2)} - \frac{1}{z} \right]$$



$$I_2 = \int \frac{2}{\pi} \sin(z) \left[\frac{1}{(z - \pi/2)} - \frac{1}{z} \right] dz$$

$$= \frac{2}{\pi} \int \frac{\sin(z)}{z - \pi/2} dz - \frac{2}{\pi} \int \frac{\sin(z)}{z} dz$$

(a) (b)

(a) $w = z - \pi/2 \Rightarrow dw = dz = \pi i e^{i\theta} d\theta$

$\hookrightarrow z = w + \pi/2$

$$\frac{2}{\pi} \int \frac{\sin(w + \pi/2)}{w} dw = \frac{2}{\pi} [2\pi i] = 4i$$

$$= \frac{2}{\pi} \int_0^{2\pi} \frac{\sin(w + \pi/2)}{\pi i e^{i\theta}} d\theta$$

(4) $= 4i$

(b) $= \frac{2}{\pi} \int \frac{\sin(z)}{z} dz = -\frac{2}{\pi} \int \frac{\sin(z)}{\pi e^{i\theta}} i\pi e^{i\theta} d\theta$

$$= -\frac{2i}{\pi} \int_0^{2\pi} \sin(z) d\theta$$

$$= -\frac{2i}{\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1} d\theta$$

$$= -\frac{2i}{\pi} \int \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \pi^{2n+1} e^{i(2n+1)\theta} d\theta \rightarrow \frac{1}{2n+1}$$

$$= -\frac{2i}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \pi^{2n+1} \frac{1}{i(2n+1)} e^{i(2n+1)\theta} \Big|_0^{2\pi}$$

$$\hookrightarrow \left[e^{i(2n+1)2\pi} - 1 \right] \frac{1}{i(2n+1)} = 0$$

$m = 2n+1 : 1, 3, \dots$

$$e^{im2\pi} = \cos(m2\pi) + i\sin(m2\pi)$$

1 0

(b) $= 0$

$$I_2 = (a) + (b) = 4i + 0$$

$\sin(z) \hookrightarrow \dots$

$$I_2 = \oint_{C_2} z(z - \pi/2) dz = 2\pi i$$

3a) Cauchy Res. Thm:

$$I_3 = \oint \frac{z^{1/2}}{1 + \sqrt{2}z + z^2} dz$$

ex: (2a) $\int \frac{\sin(z + \pi/2)}{z} dz$

* Res thm states $= 2\pi i (\text{Res}(f))$

$$f(z) = \sum_{n=-\infty}^{\infty} (z - z_0)^n \frac{\alpha_n}{n!} = \frac{\alpha_{-1}}{z} + \frac{\alpha_{-2}}{z^2} + \dots + z + z^2$$

$$\text{Res}(f) = \alpha_{-1}$$

$$\hookrightarrow \frac{\sin(z + \pi/2)}{z} = \frac{\cos(z)}{z} = \frac{1}{z} \sum_{n=0}^{\infty} \frac{z^{2n} (-1)^n}{(2n)!}$$

$$= \frac{1}{z} \left[1 + z^2 + \dots \right]$$

$$\hookrightarrow \text{Res}\left(\frac{\sin(z + \pi/2)}{z}\right) = 1$$

$$I_1 = 2\pi i * 1 = 2\pi i$$

(I₂) $\frac{\sin(z)}{z(z - \pi/2)}$ $\rightarrow z = 0$
 $z = \pi/2$

$$= 2\pi i \sum \text{Res} = 2\pi i [\text{Res}(z=0) + \text{Res}(z=\pi/2)]$$

Idea:

Res(z=0): "Remove z, and replace by its value"

$$\hookrightarrow \left(\frac{\sin(z)}{z(z - \pi/2)} \right) \cdot z = \frac{\sin(z)}{z - \pi/2} \Big|_{z=0} = 0$$

$$I_2 = 2\pi i (\text{Res}(z=0) + \text{Res}(z=\pi/2))$$

$$\text{Res}(z^{\pi/2}) =$$

$$\frac{\sin(z)}{z} \Big|_{z=\pi/2} = \frac{2}{\pi}$$

$$I_2 = 2\pi i \left(0 + \frac{2}{\pi} \right) = \underline{\underline{4i}}$$

$$\oint_{C_2} \frac{z^{1/2}}{1 + \sqrt{2}z + z^2} dz$$

$$1 + \sqrt{2}z + z^2 = 0$$

Factor:

$$z = \frac{-\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{-\sqrt{2} \pm i\sqrt{2}}{2}$$

$$= \frac{\sqrt{2}}{2} (-1 \pm i) = \frac{1}{\sqrt{2}} (-1 \pm i)$$

$$\oint \frac{z^{1/2}}{\left(z - \frac{1}{\sqrt{2}}(-1+i)\right)\left(z - \frac{1}{\sqrt{2}}(-1-i)\right)} dz$$

